

n8n - AI Automations



SoftUni Team
Technical Trainers



SoftUni



Software University

<https://about.softuni.bg>

Have Questions?

sli.do

#AI-Dev

Curious about the Final Exam?

- Check the **Regular Exam** section to find:
 - Full Project Description
 - All Requirements
 - Final Upload Deadline:
 - must submit no later than **16:00 h, 15 September 2026**

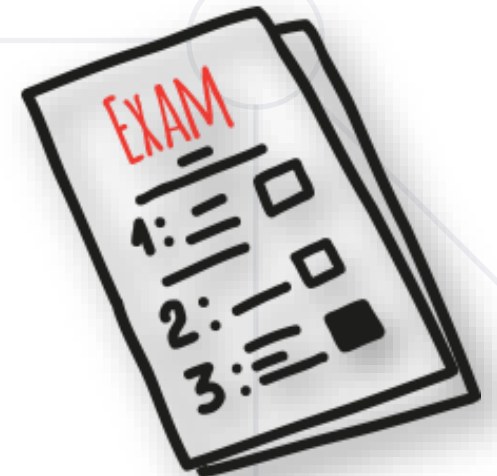


Table of Contents

1. Introduction to n8n
2. Setup workflows
3. API Integrations
4. Automating tasks
5. Advanced features





Introduction to n8n

Automation platform

- **Open, extensible automation / workflow** platform
- **Hybrid audience**: no-code UI with low-code where it helps
- Build on top of a numerous tools and APIs like **LangChain** and **LangGraph**
- Supports over a **1000 integrated functions** to 3rd party services
- Supports **complex agent structure**
- Use cases: ops automation, data pipelines, alerts, back-office glue

- N8N has 3 variants for using:
 - Self-hosted (Free) version (working on personal machine or a cloud)
 - Paid version (working on the n8n cloud)
 - Enterprise (self-hosted, extended functionality, working on personal cloud)



- Recommended way to deploy N8N on personal cloud
 - Make an account in Digital Ocean
 - Add payment card
 - Create -> Droplet
 - Choose an image -> Marketplace -> n8n
 - Choose region -> Frankfurt
 - Size, CPU option -> The cheapest is a good start
 - Link for \$200 free DigitalOcean credit -
<https://m.do.co/c/0ef4d7193b65>



N8N Dashboard & Navigation

 n8n

+

Overview

Personal

Shared with you

Templates

Variables

Insights

Help

What's New

Martin Kuvand...

Overview

All the workflows, credentials and executions you have access to

Create Workflow

Prod. executions Last 7 days 3,015 ▼9.62%	Failed prod. executions Last 7 days 25 ▼70.24%	Failure rate Last 7 days 0.8% ▼1.7pp	Time saved Last 7 days 9h ▼5h	Run time (avg.) Last 7 days 2.81s ▼0.1s
--	---	---	--	--

Workflows

Credentials

Executions

Search

Sort by last updated

My workflow 20
Last updated 2 hours ago | Created 25 August

Martin Ivanov

Inactive

RSS triggers
Last updated 1 day ago | Created 17 February

Personal

Active

AI-Website-Interested-Users
Last updated 1 day ago | Created 7 May

Personal / Website

Active

New Contact Flow
Last updated 1 day ago | Created 13 June

Personal / Website

Active

Webinar Sign Up
Last updated 1 day ago | Created 21 April

Personal / Website

Active

404 fixer
Last updated 6 days ago | Created 12 August

Personal / Website

Inactive

Exmaple-Order-List
Last updated 6 days ago | Created 3 September

Personal

Active

9

Key Concepts

- **Workflow**: a directed graph of nodes
- **Node**: a unit of work (Trigger, Transformation, Action)
- **Items**: array of records moving through nodes (item.json)
- **Credentials**: secure storage for API keys/OAuth
- **Expressions**: inline JavaScript for dynamic values
- **Executions**: run history with inputs/outputs & logs

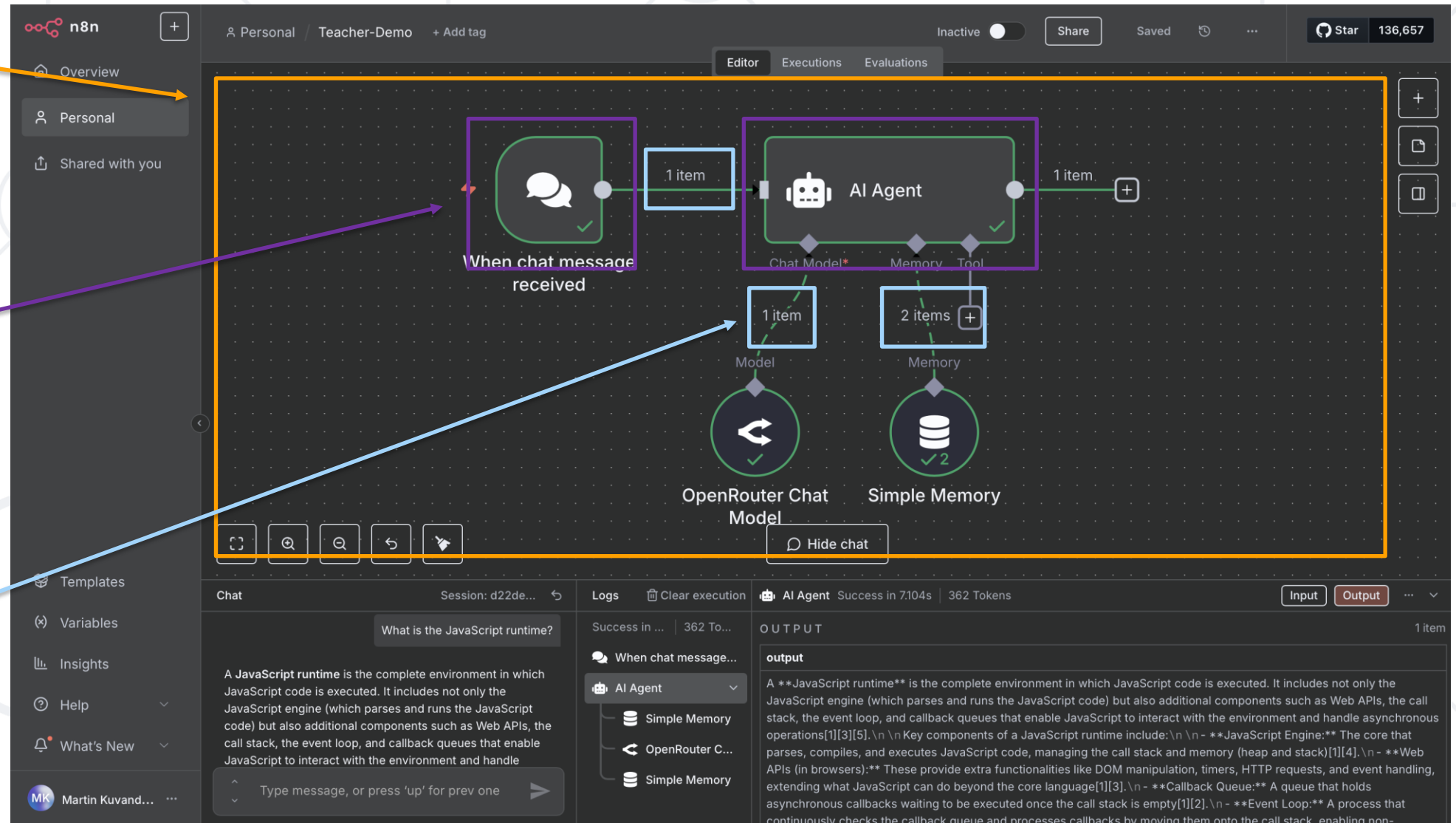


Key Concepts

Workflow

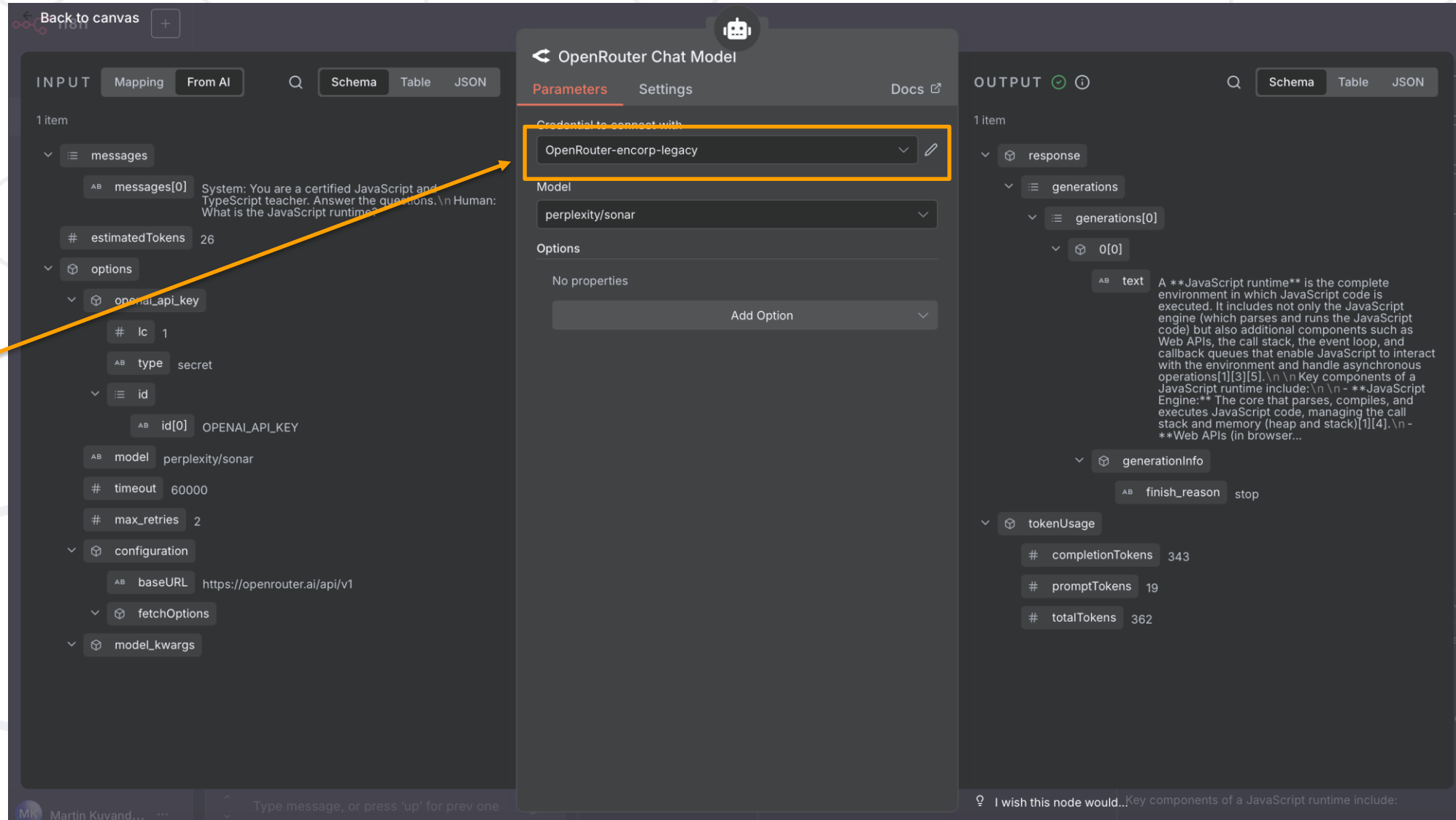
Node

Items



Key Concepts

Credentials



The screenshot displays the OpenRouter Chat Model configuration interface. The 'INPUT' section on the left shows a message history and various options, including an 'openai_api_key' field. The 'OUTPUT' section on the right shows the generated response, which is a detailed explanation of a JavaScript runtime. The 'Parameters' section in the center is highlighted with an orange box, and an orange arrow points from the 'Credentials' text on the left to the 'Credentials' field in the 'Parameters' section.

INPUT Mapping From AI Schema Table JSON

1 item

- messages
 - messages[0] System: You are a certified JavaScript and TypeScript teacher. Answer the questions. \n Human: What is the JavaScript runtime?
- estimatedTokens 26
- options
 - openai_api_key
 - lc 1
 - type secret
 - id
 - id[0] OPENAI_API_KEY
 - model perplexity/sonar
 - timeout 60000
 - max_retries 2
- configuration
 - baseURL https://openrouter.ai/api/v1
- fetchOptions
- model_kwargs

OpenRouter Chat Model Parameters Settings Docs

Credential to connect with

OpenRouter-encorp-legacy

Model

perplexity/sonar

Options

No properties

Add Option

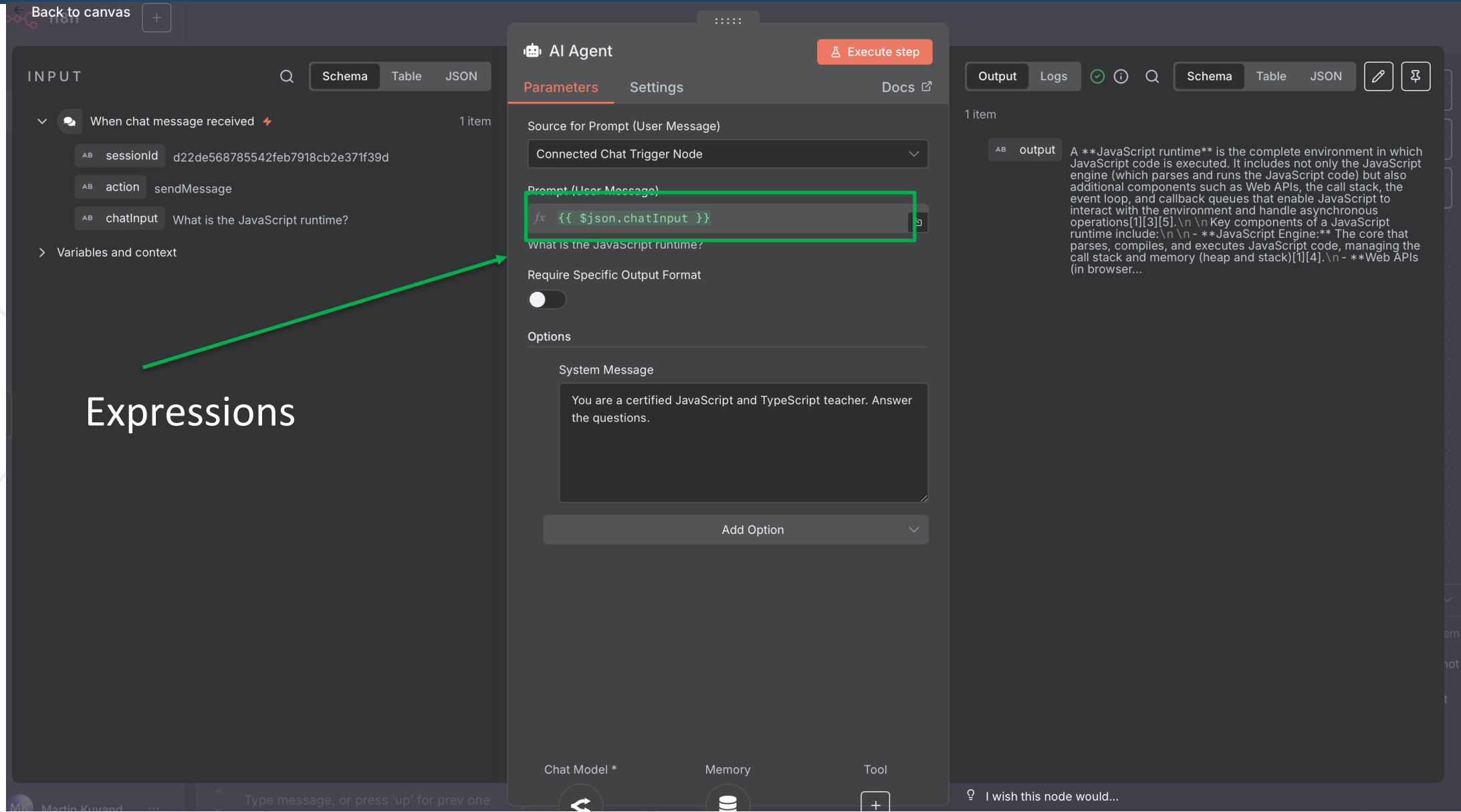
OUTPUT Schema Table JSON

1 item

- response
 - generations
 - generations[0]
 - 0[0]
 - text A **JavaScript runtime** is the complete environment in which JavaScript code is executed. It includes not only the JavaScript engine (which parses and runs the JavaScript code) but also additional components such as Web APIs, the call stack, the event loop, and callback queues that enable JavaScript to interact with the environment and handle asynchronous operations[1][3][5]. \n \n Key components of a JavaScript runtime include: \n \n - **JavaScript Engine:** The core that parses, compiles, and executes JavaScript code, managing the call stack and memory (heap and stack)[1][4]. \n - **Web APIs (in browser...
 - generationInfo
 - finish_reason stop
 - tokenUsage
 - completionTokens 343
 - promptTokens 19
 - totalTokens 362

I wish this node would...Key components of a JavaScript runtime include:

Key Concepts



The screenshot displays the 'AI Agent' configuration window, specifically the 'Parameters' tab. The 'Prompt (User Message)' field is highlighted with a green box and contains the expression `{{ $json.chatInput }}`. A green arrow points from the word 'Expressions' to this field. The 'Input' section on the left shows a chat message received with session ID `d22de568785542feb7918cb2e371f39d` and action `sendMessage`. The 'Output' section on the right shows the result of the prompt: `A **JavaScript runtime** is the complete environment in which JavaScript code is executed. It includes not only the JavaScript engine (which parses and runs the JavaScript code) but also additional components such as Web APIs, the call stack, the event loop, and callback queues that enable JavaScript to interact with the environment and handle asynchronous operations[1][3][5].\n\nKey components of a JavaScript runtime include:\n\n- **JavaScript Engine:** The core that parses, compiles, and executes JavaScript code, managing the call stack and memory (heap and stack)[1][4].\n- **Web APIs (in browser...)`.

Expressions

Key Concepts

Personal / Teacher-Demo + Add tag Inactive Share Saved Star 136,657

Editor Executions Evaluations

Sep 10, 20:39:07 Succeeded in 7.116s | ID#178017 + Add tag

Copy to editor

Executions

Auto refresh

Sep 10, 20:39:07 Succeeded in 7.116s

When chat message received

1 item

AI Agent

Chat Model*

Memory

Tool

1 item

2 items

Mode ...

Memory

OpenRouter Chat Model

Simple Memory

14

Chat

What is the JavaScript runtime?

A JavaScript runtime is the complete environment in which JavaScript code is executed. It includes not only the JavaScript engine (which parses and runs the JavaScript code) but also the standard library and the Node.js runtime.

Logs

Success... | 362 ...

When chat ...

AI Agent

AI Agent Success in 7.104s | 362 Tokens

Input Output

OUTPUT

output

A **JavaScript runtime** is the complete environment in which JavaScript code is executed. It includes not only the JavaScript engine (which parses and runs the JavaScript code) but also the standard library and the Node.js runtime.

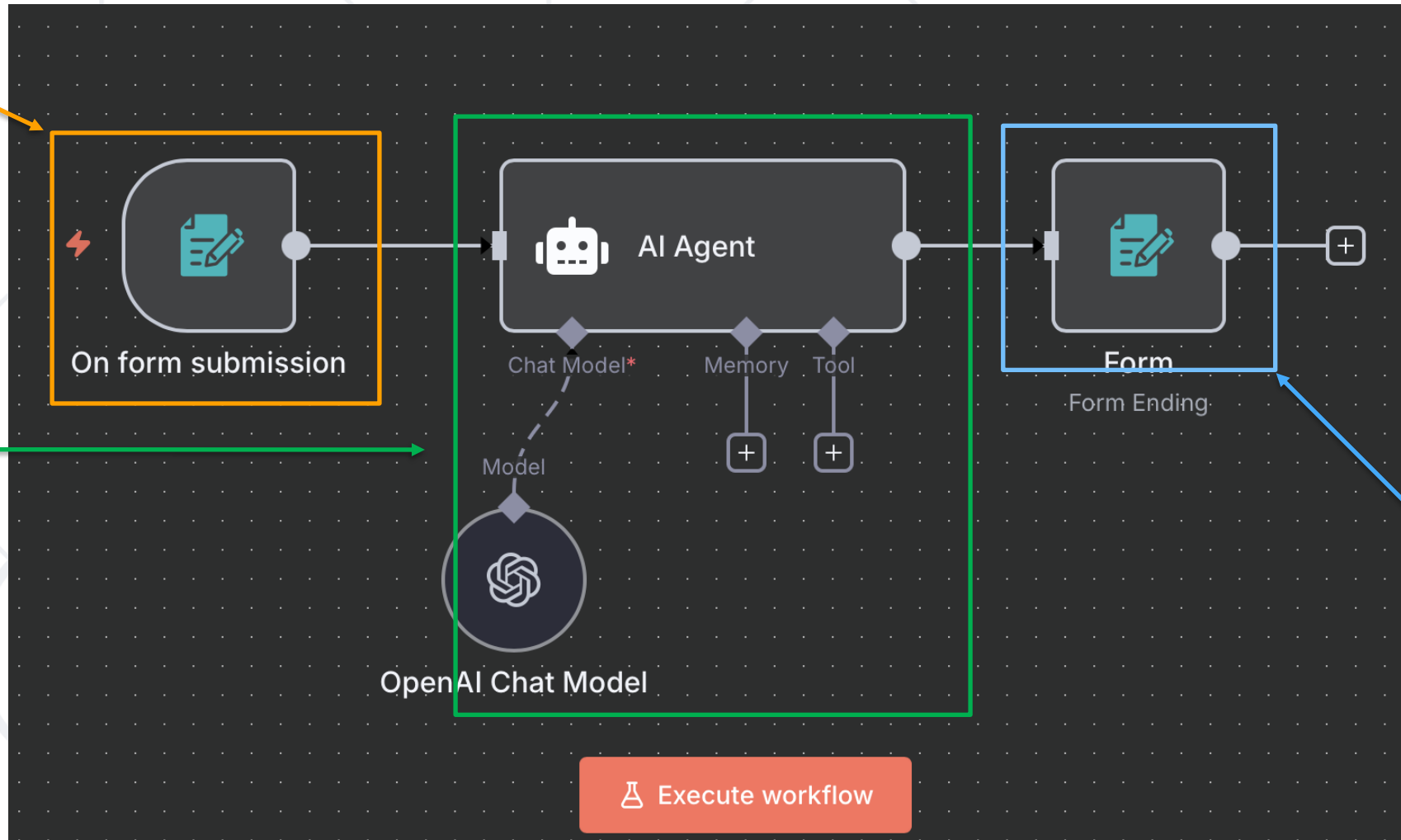
Executions

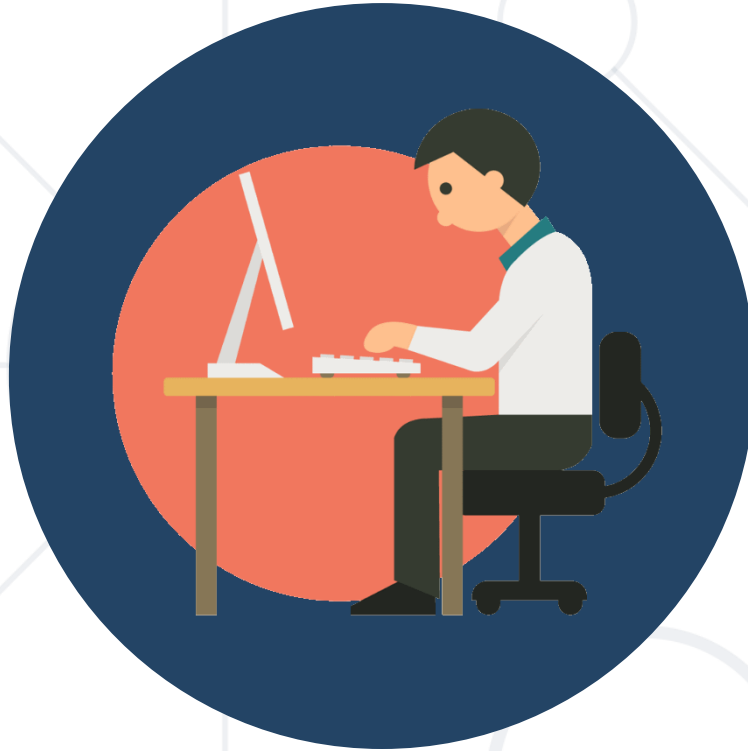
Workflow Elements

Trigger

Processing

Result





Our First Flow

Demo 1

Demo 1

- Create a workflow that **every day at 9 AM** sends an inspirational email to your email
- The workflow should use ChatGPT for the AI operations
- The workflow should use Gmail for sending emails





Building Blocks

n8n Functionalities

Note

- For the following demos I will be using a manual trigger with preset information as follows:



When clicking 'Execute workfl...' [Execute step](#)

Parameters Settings Docs

This node is where the workflow execution starts (when you click the 'test' button on the canvas).

Explore other ways to trigger your workflow (e.g on a schedule, or a webhook)

This node does not have any parameters

This data is pinned for test executions. [Unpin](#)

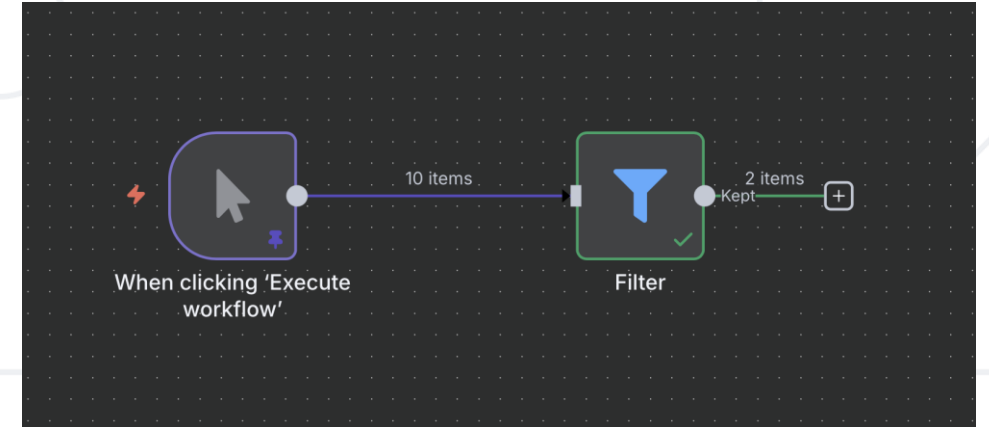
OUTPUT [Schema](#) [Table](#) JS

10 items

city	country	population
Tokyo	Japan	37400068
Cairo	Egypt	10230350
Ottawa	Canada	1076000
Canberra	Australia	472000
Nairobi	Kenya	5545000
Buenos Aires	Argentina	3075646
Berlin	Germany	3645000
Pretoria	South Africa	741651
Cape Town	South Africa	4337000
Sofia	Bulgaria	1241675

Filter node

- Filters the information based on a condition



INPUT | Schema | Table | JSON

When clicking 'Execute workfl...' | 10 items

city	country	population
Tokyo	Japan	37400068
Cairo	Egypt	10230350
Ottawa	Canada	1076000
Canberra	Australia	472000
Nairobi	Kenya	5545000
Buenos Aires	Argentina	3075646
Berlin	Germany	3645000
Pretoria	South Africa	741651
Cape Town	South Africa	4337000
Sofia	Bulgaria	1241675

Filter | Parameters | Settings | Docs

Conditions

fx `{{ $json.country }}` | is equal to | South Africa

Add condition

Convert types where required ☐

Options

No properties

Add option

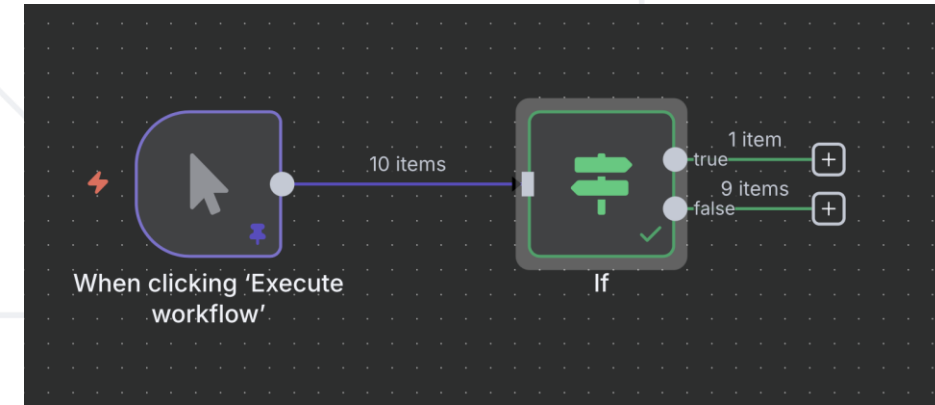
OUTPUT | Schema | Table | JSON

Kept (2 items) | Discarded (8 items)

city	country	population
Pretoria	South Africa	741651
Cape Town	South Africa	4337000

If node

- Standard if condition node with results for true and false



If

Execute step

Parameters

Settings

Docs

Conditions

fx

{{ \$json.city }}

is equal to

Tokyo

Add condition

Convert types where required

Fixed Expression

Options

No properties

Add option

OUTPUT

True Branch (1 item)

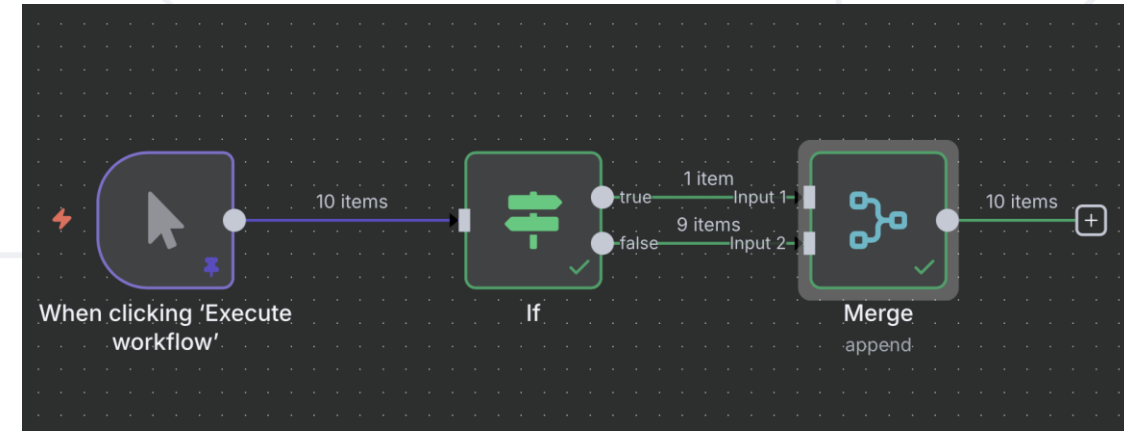
False Branch (9 items)

city	country	population
Tokyo	Japan	37400068



Merge

- Merges 2 branches into one



INPUT

Schema Table JSON

If

True Branch (1 item) False Branch (9 items)

city	country	population
Tokyo	Japan	37400068

Merge

Parameters Settings Docs

Mode

Append

Number of Inputs

2

OUTPUT

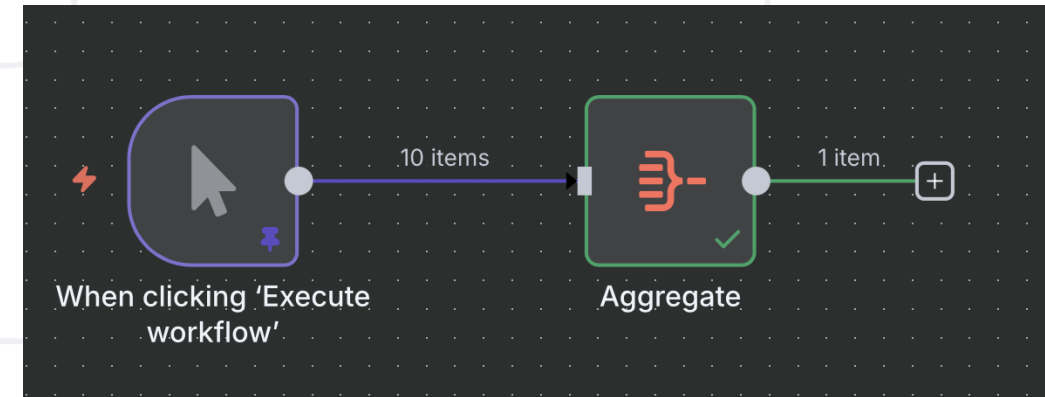
Schema Table JSON

10 items

city	country	population
Tokyo	Japan	37400068
Cairo	Egypt	10230350
Ottawa	Canada	1076000
Canberra	Australia	472000
Nairobi	Kenya	5545000
Buenos Aires	Argentina	3075646
Berlin	Germany	3645000
Pretoria	South Africa	741651
Cape Town	South Africa	4337000
Sofia	Bulgaria	1241675

Aggregate

- Aggregates multiple items into one



INPUT Schema Table JSON

When clicking 'Execute workflow' 10 items

city	country	population
Tokyo	Japan	37400068
Cairo	Egypt	10230350
Ottawa	Canada	1076000
Canberra	Australia	472000
Nairobi	Kenya	5545000
Buenos Aires	Argentina	3075646
Berlin	Germany	3645000
Pretoria	South Africa	741651
Cape Town	South Africa	4337000
Sofia	Bulgaria	1241675

Aggregate Execute step

Parameters **Settings** Docs

Aggregate: All Item Data (Into a Single List)

Put Output in Field: data

Include: All Fields

Options: No properties Add Field

OUTPUT Schema Table JSON

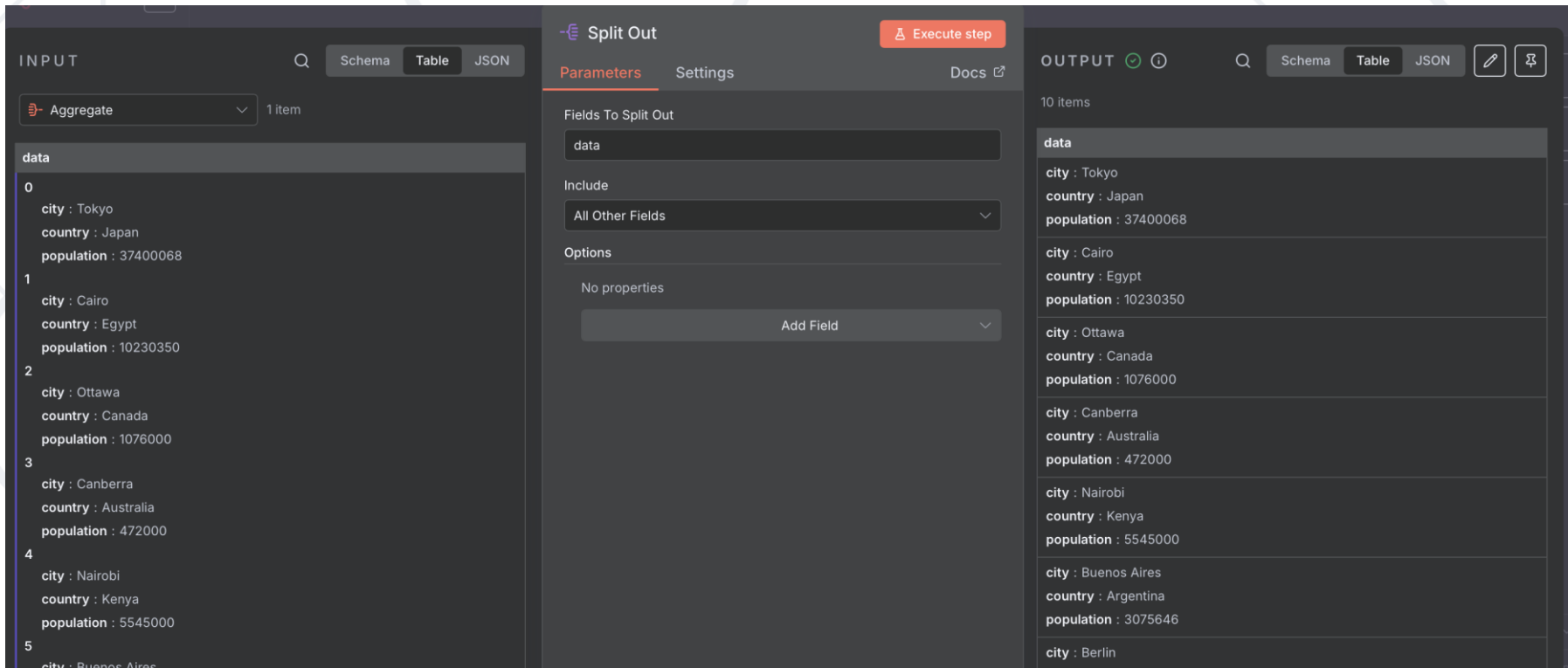
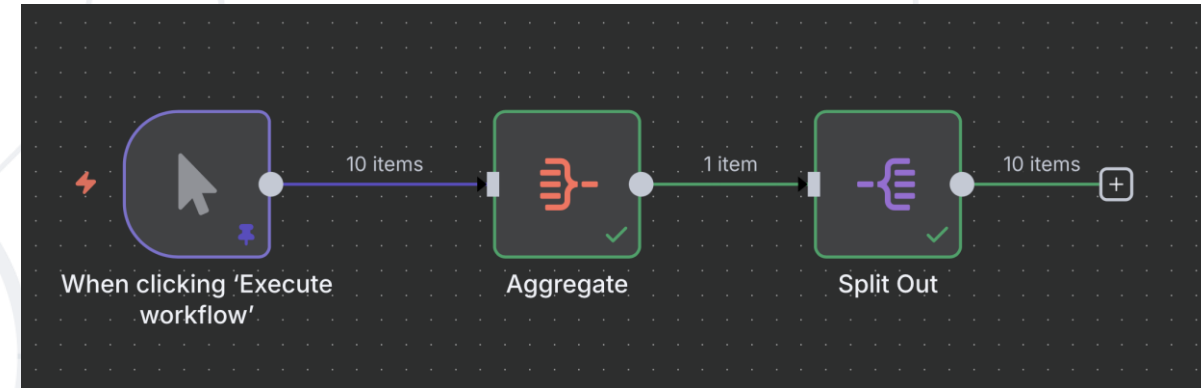
1 item

data

```
0
  city : Tokyo
  country : Japan
  population : 37400068
1
  city : Cairo
  country : Egypt
  population : 10230350
2
  city : Ottawa
  country : Canada
  population : 1076000
3
  city : Canberra
  country : Australia
  population : 472000
4
  city : Nairobi
  country : Kenya
  population : 5545000
5
  city : Buenos Aires
```

Split out

- Splits one item into many



The screenshot shows the configuration interface for the "Split Out" step. The left pane shows the input data as a table with 6 rows. The middle pane shows the configuration for the "Split Out" step, with "Fields To Split Out" set to "data" and "Include" set to "All Other Fields". The right pane shows the output data as a table with 6 rows.

INPUT

	city	country	population
0	Tokyo	Japan	37400068
1	Cairo	Egypt	10230350
2	Ottawa	Canada	1076000
3	Canberra	Australia	472000
4	Nairobi	Kenya	5545000
5	Buenos Aires	Argentina	3075646

Split Out

Parameters

Fields To Split Out

data

Include

All Other Fields

Options

No properties

Add Field

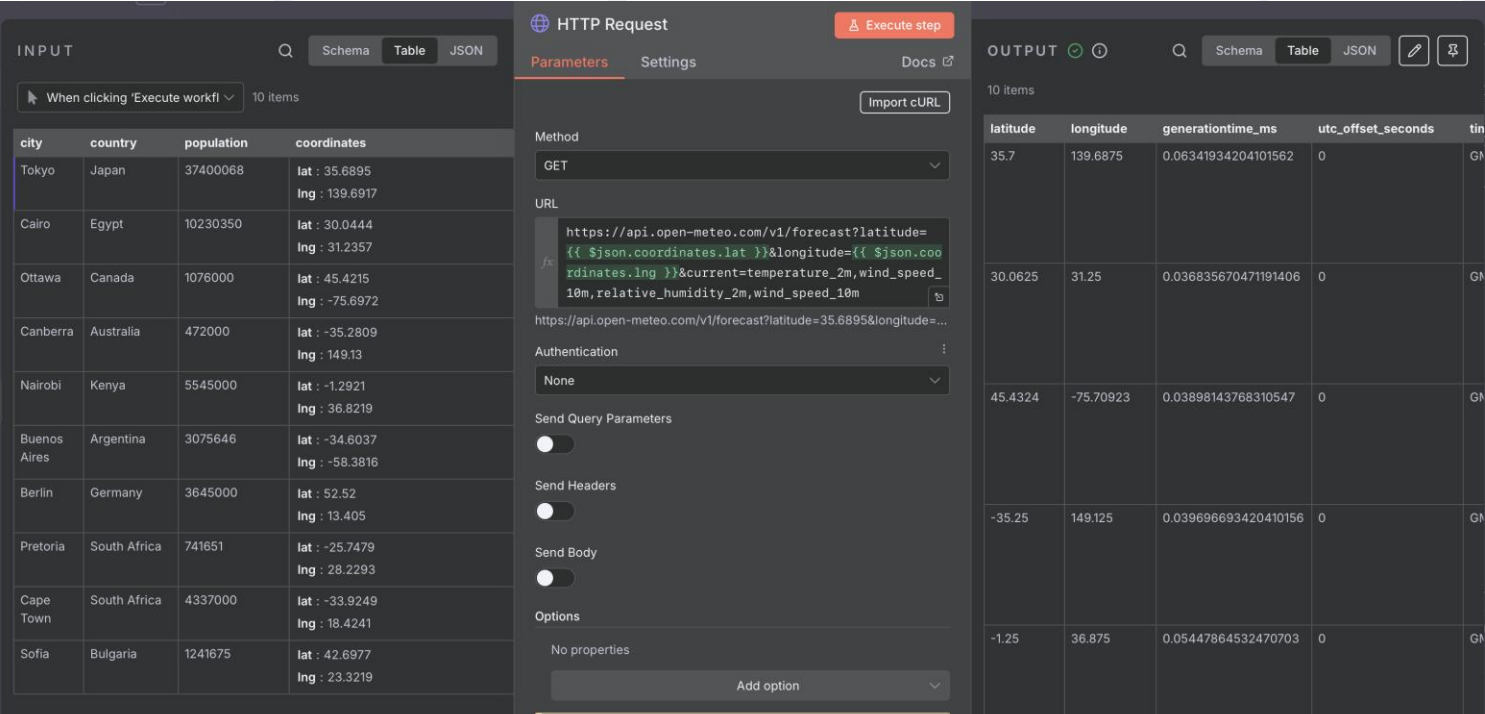
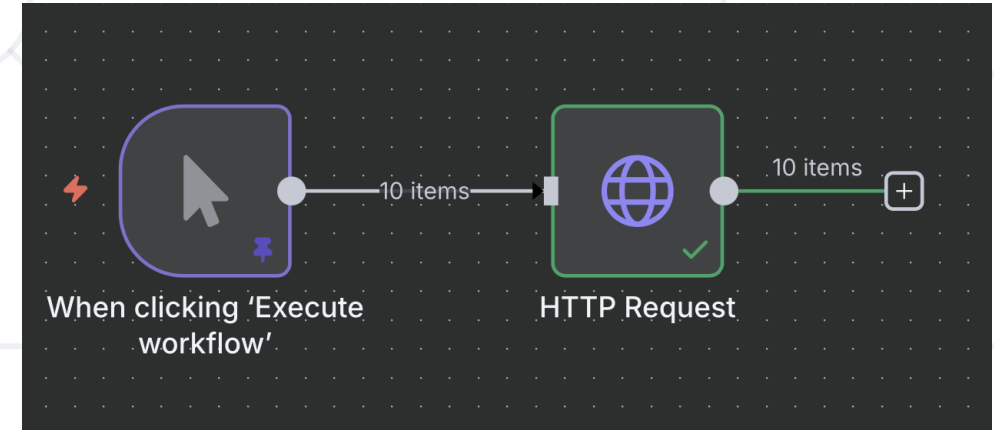
OUTPUT

	city	country	population
0	Tokyo	Japan	37400068
1	Cairo	Egypt	10230350
2	Ottawa	Canada	1076000
3	Canberra	Australia	472000
4	Nairobi	Kenya	5545000
5	Buenos Aires	Argentina	3075646



HTTP Request

- Sends an HTTP request to a server



INPUT

city	country	population	coordinates
Tokyo	Japan	37400068	lat : 35.6895 lng : 139.6917
Cairo	Egypt	10230350	lat : 30.0444 lng : 31.2357
Ottawa	Canada	1076000	lat : 45.4215 lng : -75.6972
Canberra	Australia	472000	lat : -35.2809 lng : 149.13
Nairobi	Kenya	5545000	lat : -1.2921 lng : 36.8219
Buenos Aires	Argentina	3075646	lat : -34.6037 lng : -58.3816
Berlin	Germany	3645000	lat : 52.52 lng : 13.405
Pretoria	South Africa	741651	lat : -25.7479 lng : 28.2293
Cape Town	South Africa	4337000	lat : -33.9249 lng : 18.4241
Sofia	Bulgaria	1241675	lat : 42.6977 lng : 23.3219

HTTP Request

Method: GET

URL: `https://api.open-meteo.com/v1/forecast?latitude={{ $json.coordinates.lat }}&longitude={{ $json.coordinates.lng }}¤t=temperature_2m,wind_speed_10m,relative_humidity_2m,wind_speed_10m`

Authentication: None

Send Query Parameters: ☐

Send Headers: ☐

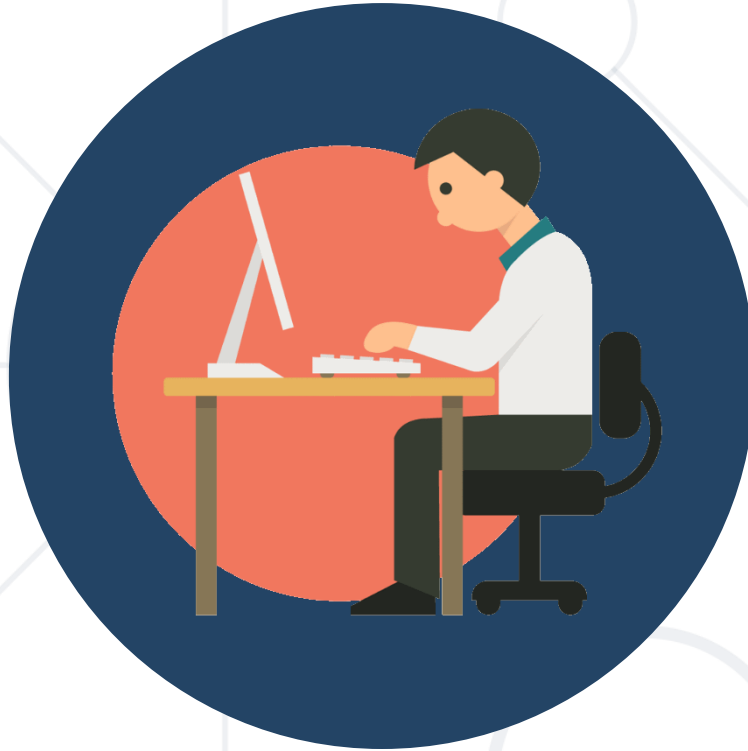
Send Body: ☐

Options: No properties

OUTPUT

latitude	longitude	generationtime_ms	utc_offset_seconds	time
35.7	139.6875	0.06341934204101562	0	GM
30.0625	31.25	0.036835670471191406	0	GM
45.4324	-75.70923	0.03898143768310547	0	GM
-35.25	149.125	0.039696693420410156	0	GM
-1.25	36.875	0.05447864532470703	0	GM





Combining all

Demo 2

Demo 2

- Create a workflow upon manual trigger gets the **weather information about 10 cities**
- After getting the information it should filter the **cities with temperatures below 15 degrees**
- Then for the cities with population above 5M it should provide only windspeed information and for cities below 5M it should provide only information about relative humidity
- The output result should be a single item

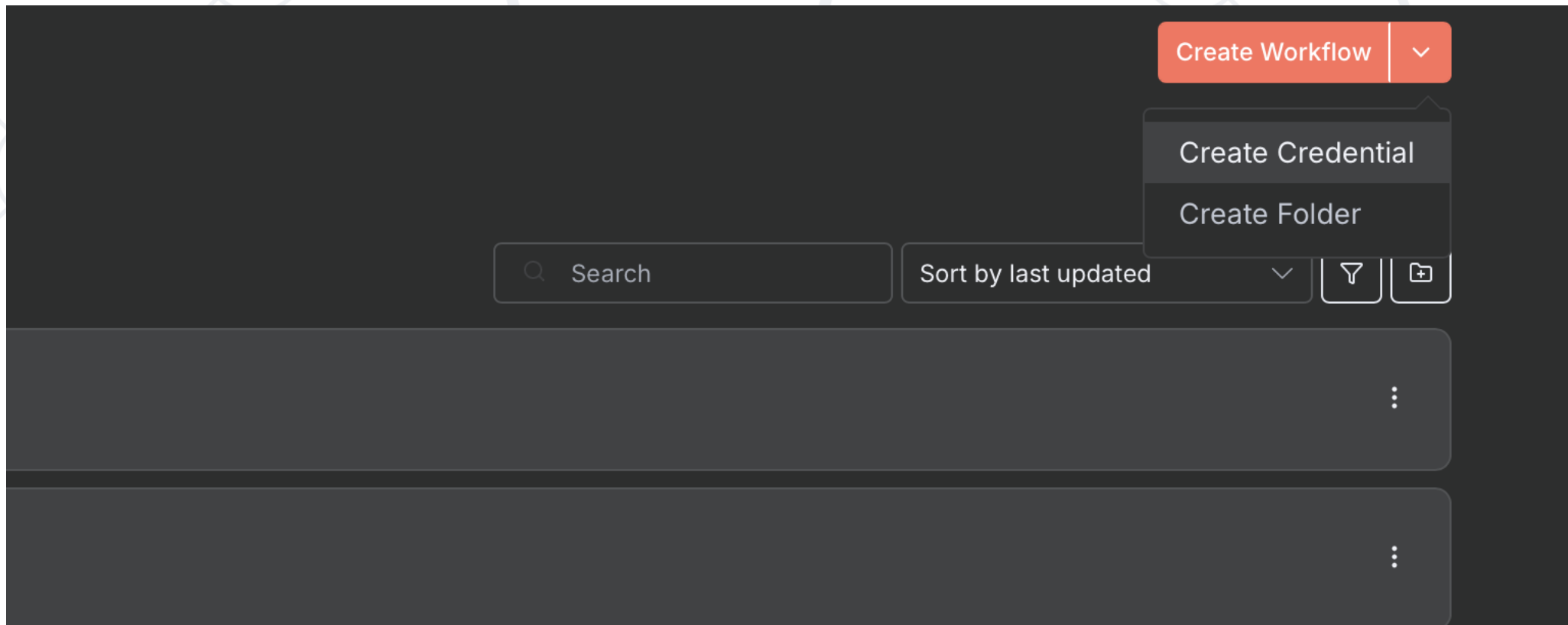




Credentials and Authentication

n8n functionalities

- Most of the integrations in n8n require **credentials** to operate. The easiest options are to just have an **API key**. The more complex systems require **OAuth2 authentication**



Common Credentials - OpenAI

- Navigate to <https://platform.openai.com/>
- Login
- Register a credit card and load a balance
- Go to API keys
- Create a new API key

Common Credentials - OpenRouter

- Navigate to <https://openrouter.ai/>
- Login / Register
- Register a credit card and load a balance
- Go to more options (The hamburger menu icon)
- Tap on Keys
- Create API Key

- Google Services -
<https://docs.n8n.io/integrations/builtin/credentials/google/oauth-single-service/#configure-your-oauth-consent-screen>
- LinkedIn -
<https://docs.n8n.io/integrations/builtin/credentials/linkedin/>
- Facebook -
<https://docs.n8n.io/integrations/builtin/credentials/facebookapp/>



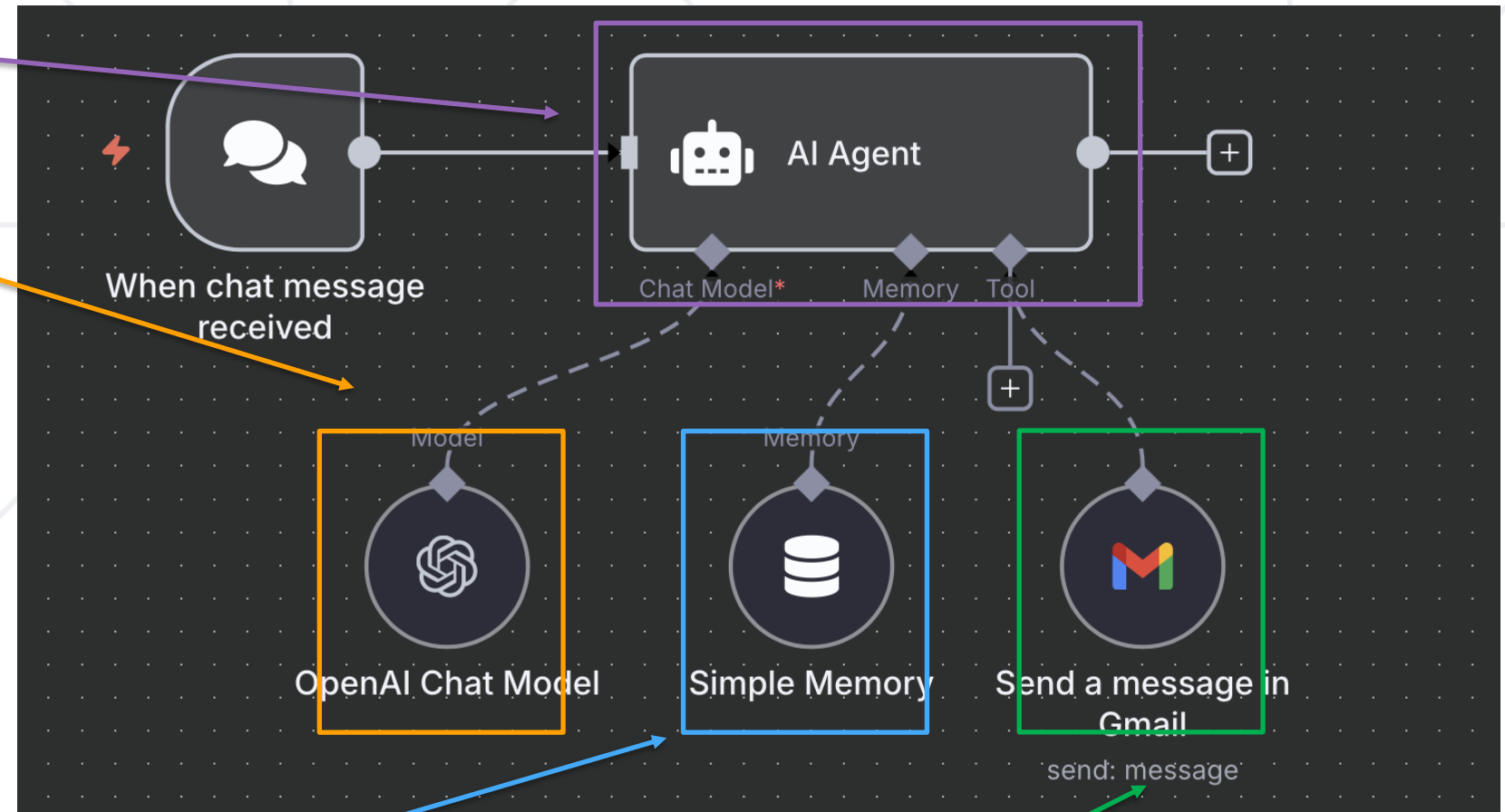
AI Blocks

n8n functionalities

Agent Structure

AI Agent node

AI Chat Model



Memory

Tools

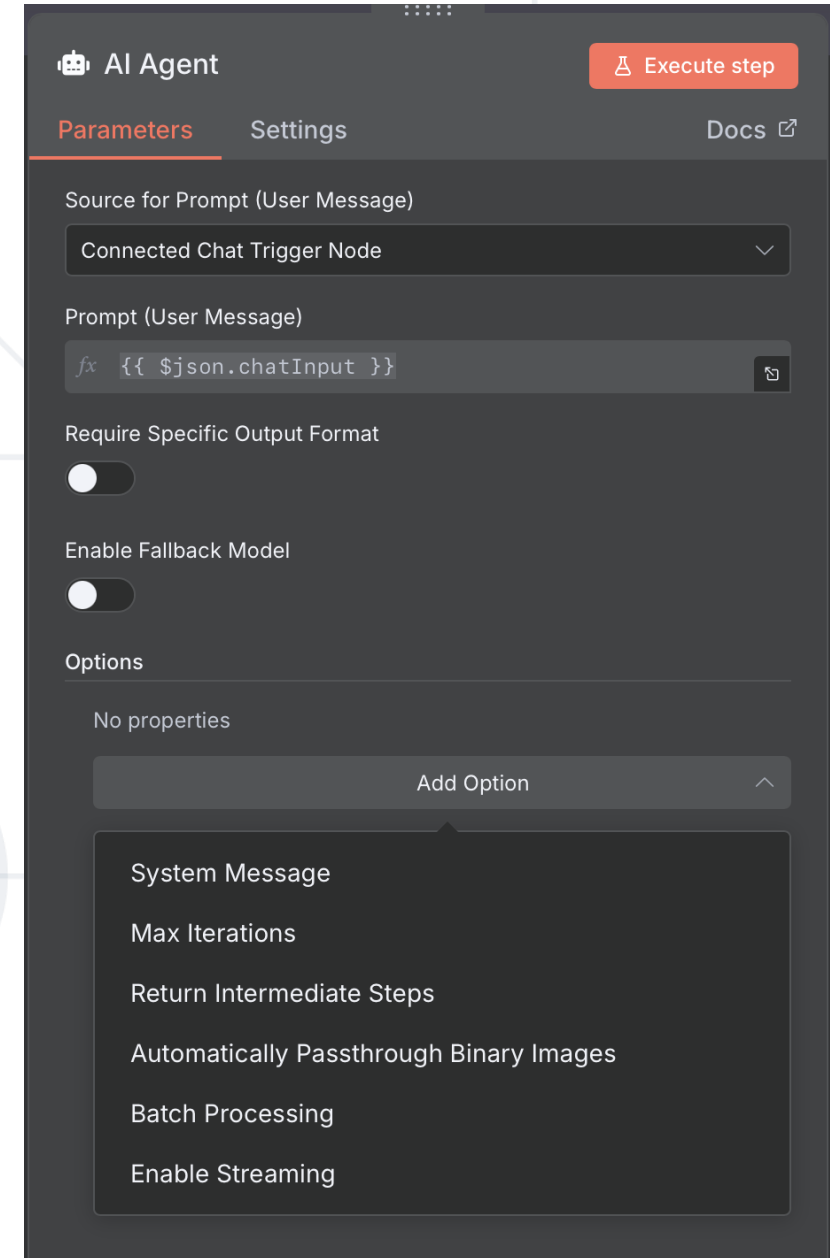
Demo 3

- Create a **personal assistant AI agent** which can read my emails, send emails and create calendar events based on a chat message



AI Agent Node Parameters

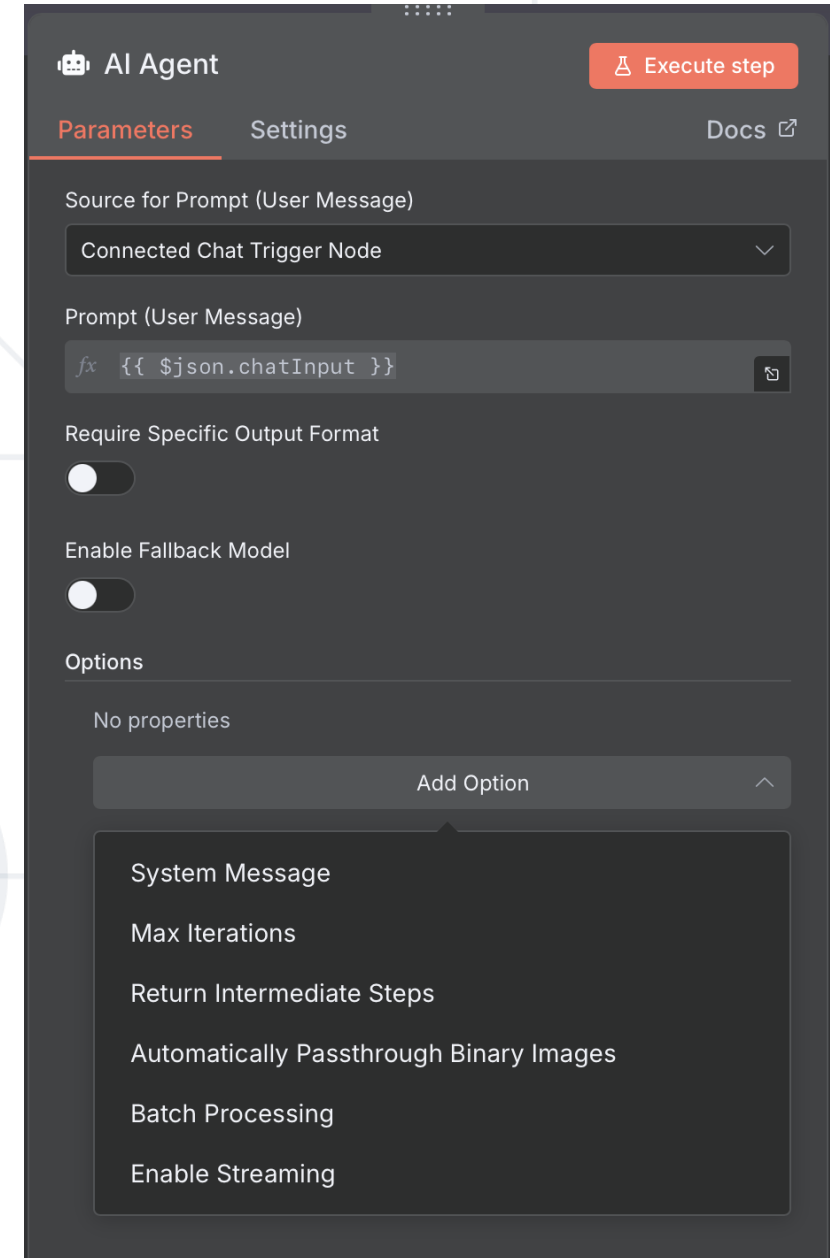
- **Require specific output format:** This is used if we want to get the result in a JSON or other specific format
- **Enable fallback model:** If the main AI model is down, it will use a backup model
- **System message:** Set a system message for the AI model
- **Max Iterations:** Sets how many times a model will be called for a single operation



The screenshot shows the configuration panel for an 'AI Agent' node. At the top right is an 'Execute step' button. Below the node name are tabs for 'Parameters' (selected) and 'Settings', along with a 'Docs' link. The 'Parameters' section includes a dropdown for 'Source for Prompt (User Message)' set to 'Connected Chat Trigger Node', a text field for 'Prompt (User Message)' containing the JSON template `{{ $json.chatInput }}`, and two toggle switches for 'Require Specific Output Format' and 'Enable Fallback Model', both currently turned off. An 'Options' section follows, showing 'No properties' and an 'Add Option' button. A dropdown menu is open, listing several options: 'System Message', 'Max Iterations', 'Return Intermediate Steps', 'Automatically Passthrough Binary Images', 'Batch Processing', and 'Enable Streaming'.

AI Agent Node Parameters

- **Return Intermediate Steps:** Will show all the actions the agent did in the output
- **Automatically passthrough binary image:** Used if you are processing images
- **Batch processing:** Will combine few agent operations into one per multiple items
- **Enable streaming:** Will be streaming the output, not just providing a single message at the end



The screenshot shows the configuration panel for an 'AI Agent' node. At the top right is an 'Execute step' button. Below the node name are tabs for 'Parameters' (selected) and 'Settings', along with a 'Docs' link. The 'Parameters' section includes a 'Source for Prompt (User Message)' dropdown set to 'Connected Chat Trigger Node', a 'Prompt (User Message)' text area containing the JSON template `{{ $json.chatInput }}`, and two toggle switches for 'Require Specific Output Format' and 'Enable Fallback Model', both currently turned off. An 'Options' section follows, showing 'No properties' and an 'Add Option' button. A dropdown menu is open, listing the following options: 'System Message', 'Max Iterations', 'Return Intermediate Steps', 'Automatically Passthrough Binary Images', 'Batch Processing', and 'Enable Streaming'.

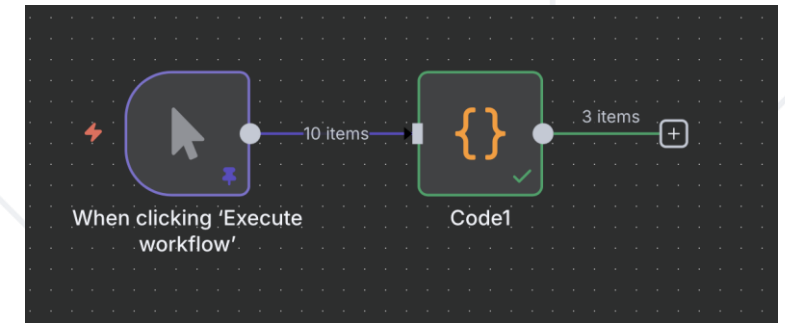


Advanced Features

n8n functionalities

Code Nodes

- N8N has the ability to execute simple js and python code



Back to canvas +

INPUT

When clicking 'Execute workflow' 10 items

```
[
  {
    "city": "Tokyo",
    "country": "Japan",
    "population": 37400068,
    "coordinates": {
      "lat": 35.6895,
      "lng": 139.6917
    }
  },
  {
    "city": "Cairo",
    "country": "Egypt",
    "population": 10230350,
    "coordinates": {
      "lat": 30.0444,
      "lng": 31.2357
    }
  },
  {
    "city": "Ottawa",
    "country": "Canada",
    "population": 1076000,
    "coordinates": {
      "lat": 45.4215,
      "lng": 75.6972
    }
  }
]
```

Code1

Parameters Settings Docs

Mode

Run Once for All Items

Language

JavaScript

JavaScript

```
1 // Loop over input items and add a new field
  called 'myNewField' to the JSON of each one
2 let results = []
3
4 for (const item of $input.all()) {
5   if (item.json.population > 5000000) {
6     results.push(item)
7   }
8 }
9
10 return results;
```

Type \$ for a list of special vars/methods. Debug by using console.log() statements and viewing their output in the browser console.

OUTPUT

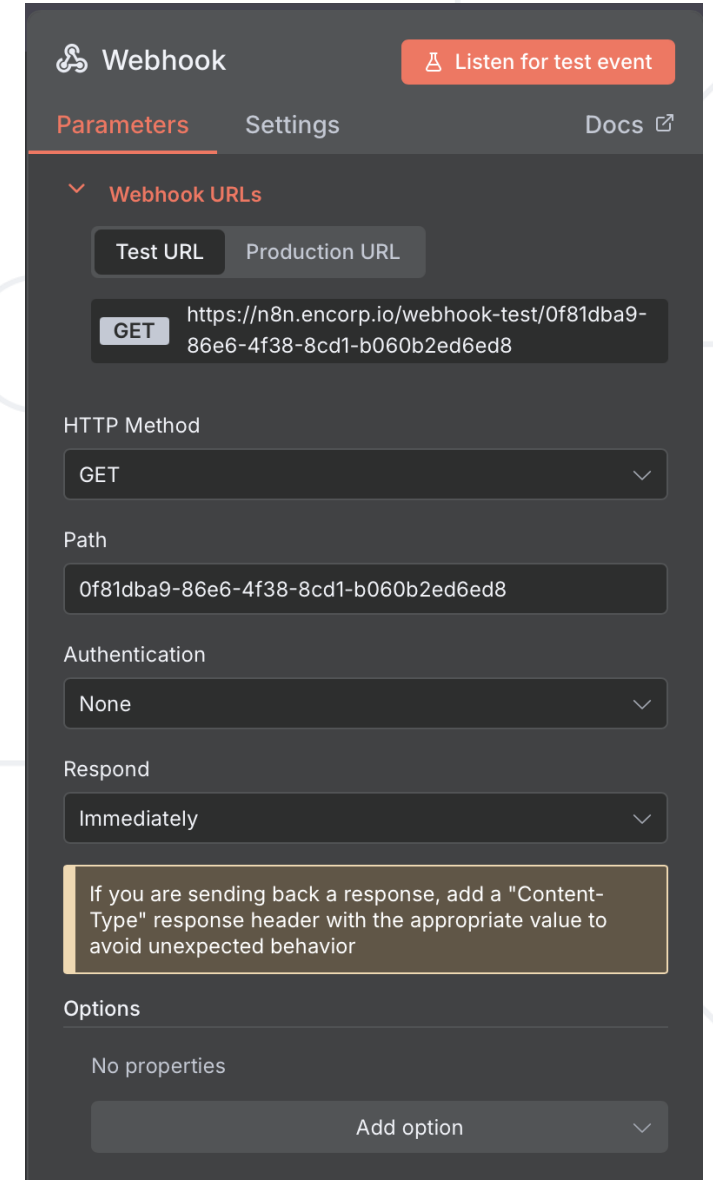
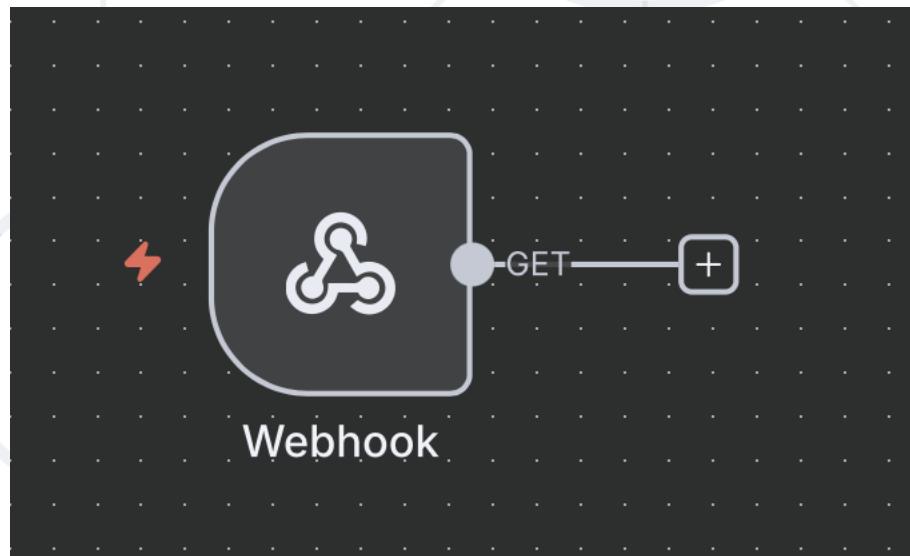
To make sure expressions after this node work, return the input items that produced each output item. [More info](#)

3 items

city	country	population	coordinates
Tokyo	Japan	37400068	lat : 35.6895 lng : 139.6917
Cairo	Egypt	10230350	lat : 30.0444 lng : 31.2357
Nairobi	Kenya	5545000	lat : -1.2921 lng : 36.8219

Webhook Triggers

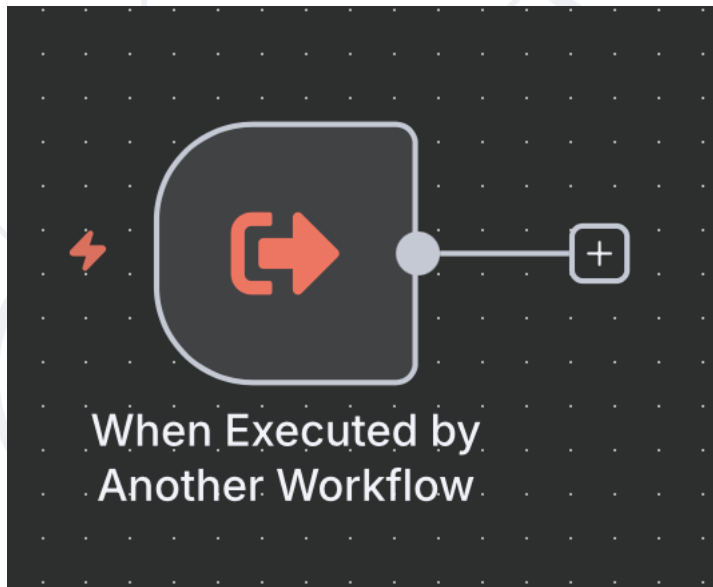
- There are integrated webhooks in n8n for incoming communication by other services



The screenshot shows the n8n Webhook configuration panel. At the top, there's a 'Webhook' header with a 'Listen for test event' button. Below this are tabs for 'Parameters' (selected), 'Settings', and 'Docs'. The 'Parameters' tab contains a section for 'Webhook URLs' with 'Test URL' and 'Production URL' buttons. The 'Test URL' is displayed as 'https://n8n.encorp.io/webhook-test/0f81dba9-86e6-4f38-8cd1-b060b2ed6ed8'. Below this, the 'HTTP Method' is set to 'GET', the 'Path' is '0f81dba9-86e6-4f38-8cd1-b060b2ed6ed8', 'Authentication' is 'None', and 'Respond' is 'Immediately'. A note states: 'If you are sending back a response, add a "Content-Type" response header with the appropriate value to avoid unexpected behavior'. The 'Options' section is currently empty with a 'No properties' label and an 'Add option' button.

Sub-workflows

- Complex workflows may require **splitting** one workflow to several smaller sub-workflows



 When Executed by Another W...  Execute step

Parameters

SettingsDocs 

Input data mode

Define using fields below 

Workflow Input Schema

Name

city

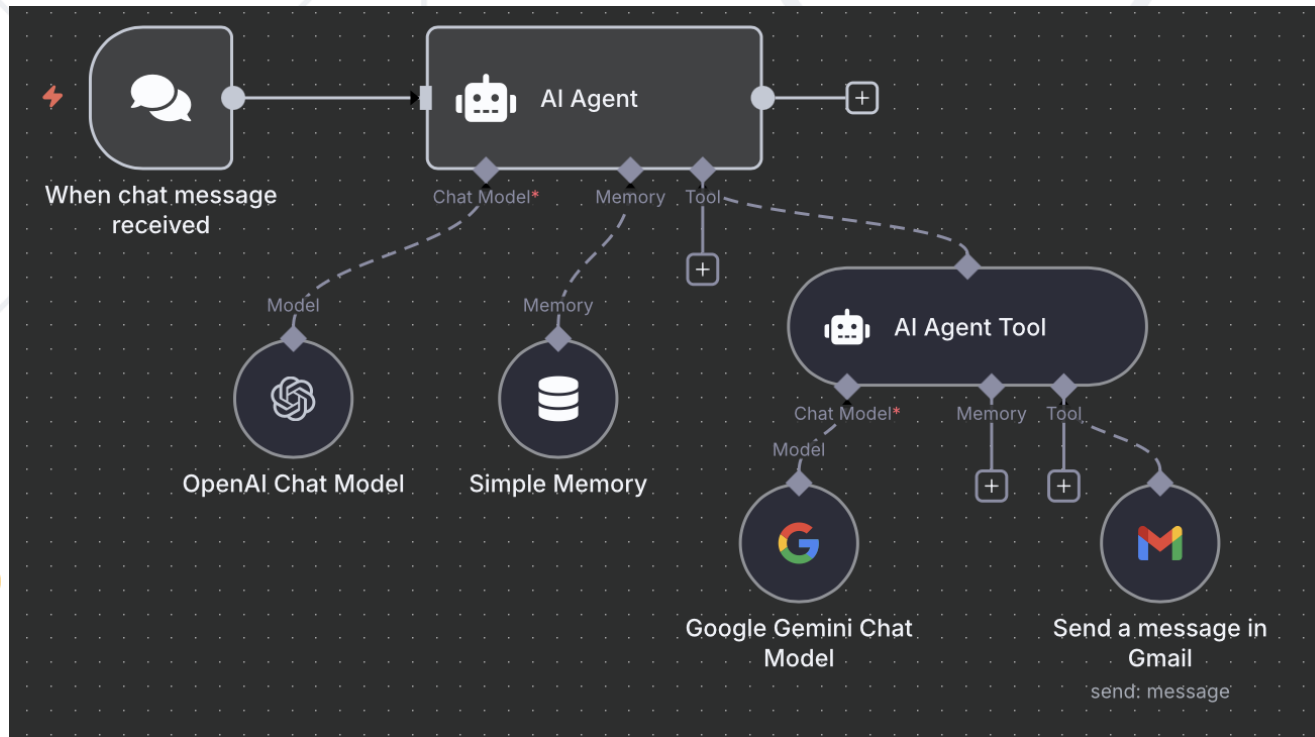
Type

String 

Add field

AI Agent Tool (Sub-Agents)

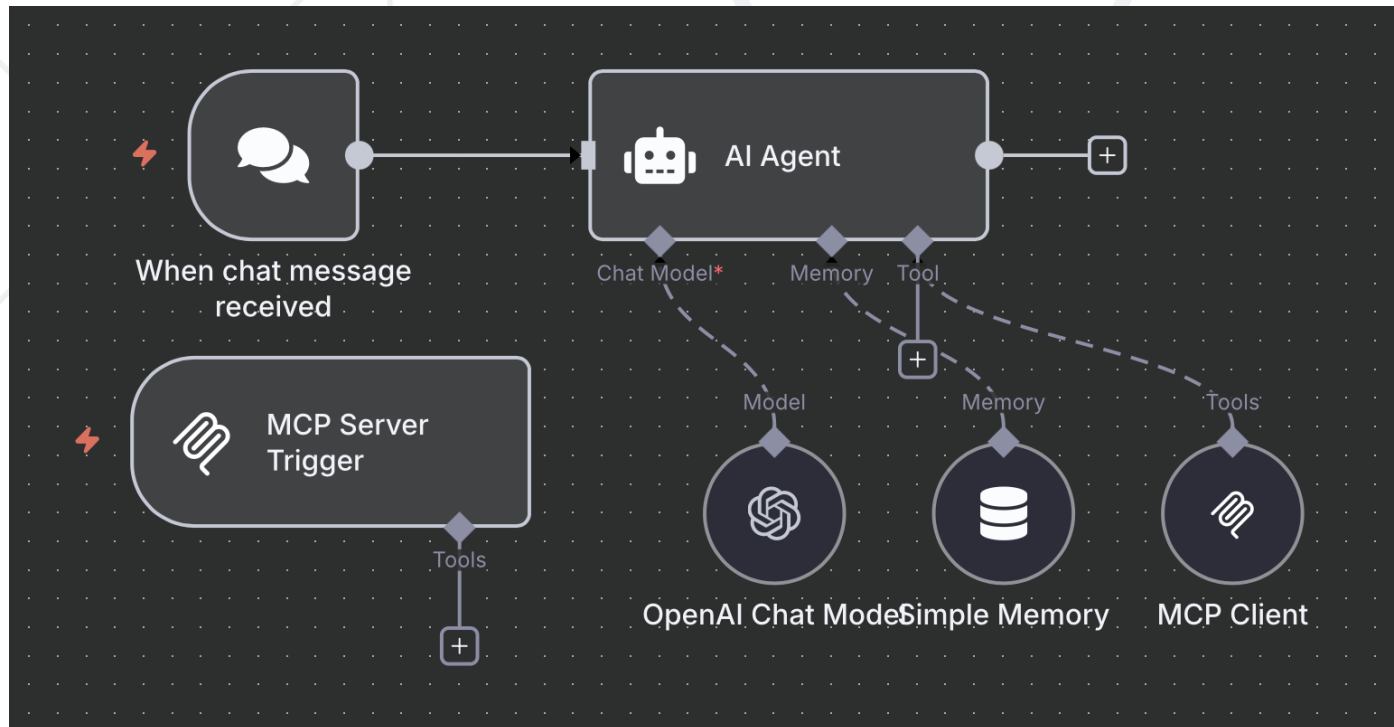
- Having more than 6 tools in a single agent may cause **hallucinations**. For complex agents, working with sub-agents is recommended



The screenshot shows the configuration interface for the "AI Agent Tool". The interface includes a header with a robot icon, the title "AI Agent Tool", and an "Execute step" button. Below the header, there are tabs for "Parameters" (selected) and "Settings", along with a "Docs" link. The "Parameters" section contains a "Description" field with the text "This is an agent that sends emails", a "Prompt (User Message)" field with the text "Defined automatically by the model", and a toggle for "Require Specific Output Format". The "Options" section shows "No properties" and an "Add Option" button.

MCP Server and Client

- n8n can operate as a both **MCP client** and **server**, enabling call to other MCPs and from other services



The screenshot shows the **MCP Client** configuration interface in n8n. The interface has a dark theme and includes a robot icon in the top right corner. The **Parameters** tab is selected, showing the following settings:

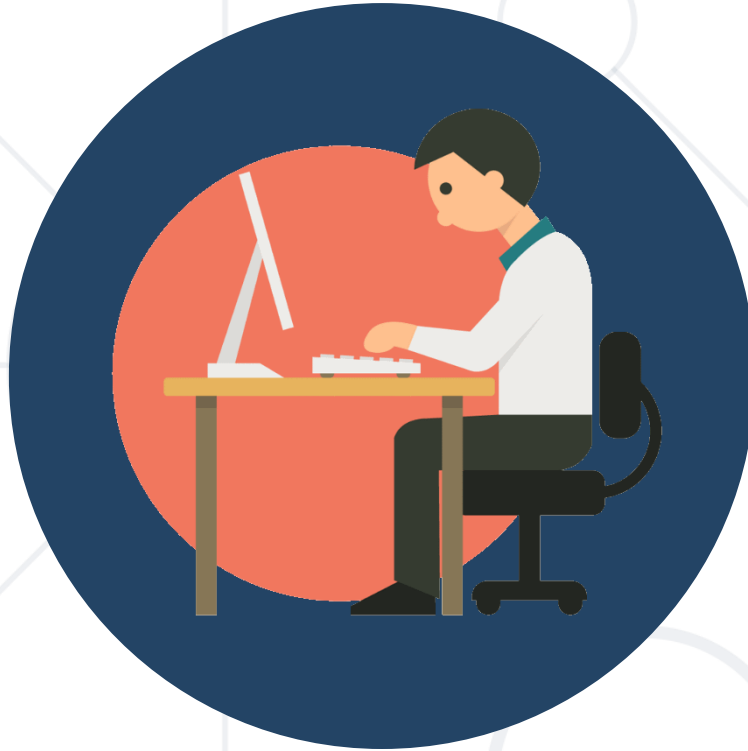
- Endpoint:** `https://mcp.myserver.com/mcp`
- Server Transport:** **Server Sent Events (Deprecated)**
- Authentication:** **None**
- Tools to Include:** **All**

There is an **Execute step** button in the top right corner and a **Docs** link in the bottom right corner.

Tips and tricks & Real-work examples



- If a workflow has **more than 20 nodes** - **split it** into multiple workflows
- Select the **right model** based on the task. Larger models usage is getting expensive when tools are invoked
- N8N is working on a **single thread**. Running big code nodes may temporarily halt the operations of n8n
- When using API calls to external services it's recommended to **use loops with wait node**



Personal assistant AI Agent

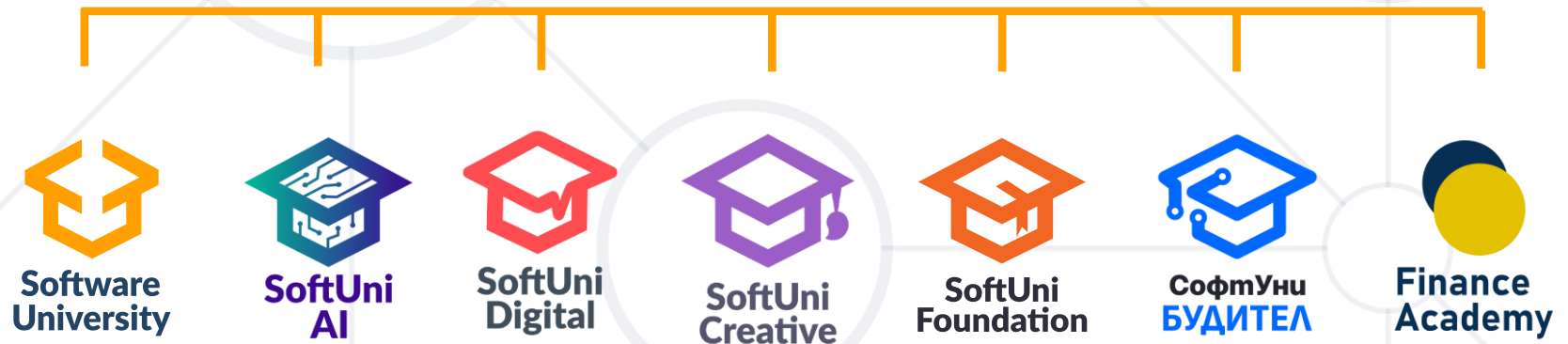
Demo 4

- You have to create a **personal assistant AI agent** which is triggered by a webhook
- Functions:
 - Search for **trending news** in the internet. Should be implemented as MCP service
 - **Fetch weather data** and **respond**, including average temperature
 - Send **emails**
 - Create **calendar events**
 - Prepares **content** for Facebook posts

Questions?



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